Likelihood and likelihood estimators

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Outline - rest of the semester

Statistical inference

- Maximum likelihood estimation
  - Maximum likelihood estimators and their properties
  - Sufficient statistics, Fisher's information, asymptotical confidence intervals
- Statistical hypothesis testing in linear models
- Nonparametrics, categorical data analysis
- Goodness of fit tests

Literature

- *DeGroot, M. H., Schervish, M. J. (n.d.). Probability and Statistics (4th ed.). Pearson.*
- *Anděl, J. (2011). Základy matematické statistiky (Vyd. 3). Matfyzpress.*
- *Zvára, K. (2019). Regrese (Vydání druhé). Matfyzpress.*

Outline - today

1. Revision of some necessary terms
   - Continuous random variable
   - Independence of random variables
   - Bayes' theorem for random variables
2. Statistical Inference
   - Definitions
   - Likelihood

Continuous random variable

Definition (Hustota pravděpodobnosti)

Probability density function (PDF) of a continuous random variable with cumulative distribution function (CDF) $F(x)$ is

$$f(x) = \frac{d}{dx}F(x)$$

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- probability of any given x is zero ($P(X = x) = 0$).
- for nonzero probability it is necessary to limit x by an interval(s)

Characteristics of a random variable

Compared to discrete random variables, probability and random variable characteristic are computed via integration instead of summation. e.g.:

Definition (Střední hodnota)

Let X be a continuous random variable. Expectation of X is

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

if the value of the integral is finite

Higher order moments

$$E[X^k] = \int_{-\infty}^{\infty} x^k f(x)\, dx$$

$$Var(X) = E[(X - E(X))^2] = E(X^2) - [E(X)]^2$$

Example - exponential distribution

$$f(x) = \lambda e^{-\lambda x} \qquad x \geq 0$$

$$F(x) = \int_0^x \lambda e^{-\lambda t}\,dt \left| \begin{array}{l} -\lambda t = a \\ -\lambda\,dt = da \\ dt = \frac{da}{-\lambda} \end{array} \right| \doteq \int -e^{a}\,da = -e^{a} \doteq \left[-e^{-\lambda t}\right]_0^x = -e^{-\lambda x} + 1 \qquad x \geq 0$$

$$\llcorner\ 0 \qquad x < 0$$

$$E(X) = \int_0^\infty x \cdot f(x)\,dx = \int_0^\infty x \cdot \lambda \cdot e^{-\lambda x}\,dx \left| \begin{array}{ll} u = x & u' = 1 \\ v' = \lambda \cdot e^{-\lambda x} & v = -e^{-\lambda x} \end{array} \right| = \overset{0}{\left[-x\,e^{-\lambda x}\right]_0^\infty} + \int_0^\infty e^{-\lambda x}\,dx =$$

$$\int_0^\infty e^{-\lambda x}\,dx = \left[-\frac{1}{\lambda} e^{-\lambda x}\right]_0^\infty = 0 + \underline{\underline{\frac{1}{\lambda}}}$$

Definition - Quantiles/Percentiles

Let X be a random variable with CDF. F. For each p strictly between 0 and 1, define $F^{-1}(p)$ to be the smallest value x such that $F(x) \geq p$. Then $F^{-1}(p)$ is called the p quantile of X or the 100p percentile of X. The function $F^{-1}(p)$ defined here on the open interval (0, 1) is called the quantile function of X.

Notation - Median, First quartile, Third quartile

$$Median = X_{0,5} = F^{-1}(0,5)$$

$$First\ quartile = Q1 = F^{-1}(0,25)$$

$$Third\ quartile = Q3 = F^{-1}(0,75)$$

Conditional PDF

Definition (Podmíněná hustota pravděpodobnosti)

Let X and Y have a continuous joint distribution with joint PDF f and respective marginals f_1 and f_2. Let y be a value such that $f_2(y) > 0$. Then the conditional PDF g_1 of X given that $Y = y$ is defined as follows:

$$g_1(x|y) = \frac{f(x,y)}{f_2(y)} \text{ for } x \in \mathbb{R}$$

For values of y such that $f_2(y) = 0$, we are free to define $g_1(x|y)$ however we wish, so long as $g_1(x|y)$ is a PDF as a function of x.

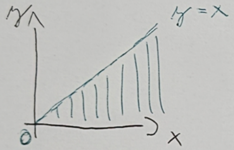
Example

A manufacturing process consists of two stages. The first stage takes Y minutes, and the whole process takes X minutes (which includes the first minutes). Suppose that X and Y have a joint continuous distribution with joint PDF

$$f(x,y) = \begin{cases} e^{-x} & \textit{for } 0 \leq y \leq x < \infty \\ 0 & \textit{otherwise}. \end{cases}$$

Calculate conditional PDF of X given Y=y.

$$f(x;y)=\begin{cases} e^{-x} & 0 \le y \le x \le \infty \\ 0 & \text{jinde} \end{cases}$$



$$f_2(y)=\int_y^\infty f(x;y)\,dx=\int_y^\infty e^{-x}\,dx=\left[-e^{-x}\right]_y^\infty=e^{-y}$$

$$g(x|Y=y)=\frac{e^{-x}}{e^{-y}}=\underline{\underline{e^{y-x}}}$$

Independent random variables

Definition (Nezávislé náhodné veličiny)

Suppose that X and Y are two random variables having a joint PMF, PDF., or PMF/PDF f . Then X and Y are independent if and only if for every value of y such that $f_2(y) > 0$ and every value of x,

$$g_1(x|y) = f_1(x)$$

Bayes' theorem for random variables

Theorem

If $f_2(y)$ is the marginal PMF. or PDF of a random variable Y and $g1(x|y)$ is the conditional PMF or PDF of X given Y $= y$, then the conditional PMF or PDF of Y given X $=$ x is

$$g_2(y|x) = \frac{g_1(x|y) f_2(y)}{f_1(x)}$$

Statistical inference

Apparatus for answering questions like:
"What is the probability of future patient will respond successfully to a treatment after we observe the result from a collection of other patients?"
"Can we assume some machine is working properly after observing its output?"

Definition

A statistical inference is a procedure that produces a probabilistic statement about some or all parts of a statistical model.

Statistical Model

Definition

A statistical model consists of an identification of random variables, the identification of any parameters of those distributions that are assumed unknown and possibly hypothetically observable, and (if desired) a specification for a (joint) distribution for the unknown parameter(s). When we treat the unknown parameter(s) θ as random, then the joint distribution of the observable random variables indexed by θ is understood as the conditional distribution of the observable random variables given θ.

Example

A company sells electronic components and they are interested in knowing as much as they can about how long each component is likely to last. They can collect data on components that have been used under typical conditions.They choose to use exponential distribution to model the length of time (in years) from when a component is put into service until it fails. They would like to model the components as all having the same failure rate λ, but there is uncertainty about the specific numerical value of λ. They hope to learn more about λ from examining the sample data $X_1, ..., X_n$. They can never learn λ precisely, because that would require observing the entire infinite sequence $X_1, X_2, ...$ For this reason, λ is only hypothetically observable.

Parameter space

Definition (Parametrický prostor)

In a problem of statistical inference, a characteristic or combination of characteristics that determine the joint distribution for the random variables of interest is called a parameter of the distribution. The set Θ of all possible values of a parameter θ or of a vector of parameters $(\theta_1, ..., \theta_k)$ is called the parameter space.

Statistic

Definition (Statistika)

Suppose that the observable independent identically distributed (IID) random variables of interest are $X_1, ..., X_n$. Let r be an arbitrary real-valued function of n real variables. Then the random variable $T = r(X_1, ..., X_n)$ is called a statistic.

Likelihood function

Definition (Věrohodnostní funkce)

When the joint PDF or the joint PMF $f_n(\mathbf{x}, \theta)$ of the observations in a random sample is regarded as a function of θ for given values of $x_1, \ldots, x_n$, it is called the likelihood function.

Notation

Likelihood function $L(\theta) = f_n(\mathbf{x}, \theta)$
Log-likelihood function $l(\theta) = \ln(L(\theta))$

Maximum likelihood estimator/estimate MLE

Definition (Maximálně věrohodný odhad)

For each possible observed vector $\mathbf{x}$, let $\delta(\mathbf{x}) \in \Theta$ denote a value of $\theta \in \Theta$ for which the likelihood function $L(\theta)$ is a maximum, and let $\hat{\theta} = \delta(\mathbf{X})$ be the estimator of θ defined in this way. The estimator $\hat{\theta}$ is called a maximum likelihood estimator of θ. After $\mathbf{X} = \mathbf{x}$ is observed, the value $\delta(\mathbf{x})$ is called a maximum likelihood estimate of θ.

The main idea is to find a parameter value that fits the observed values "the best" without implementing any other information.

Example - MLE of exponential distribution

$$X_i \sim f(x) = \begin{cases} \lambda \cdot e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad \Bigg| \quad n \dots X_1, \dots, X_n \text{ I.I.D.}$$

$$L(\lambda) = \prod_{i=1}^{n} f(x_i) = \prod_{i=1}^{n} \left(\lambda \cdot e^{-\lambda x_i}\right) = \lambda^{n} \cdot e^{-\sum_{i=1}^{n} \lambda x_i}$$

$$\ell(\lambda) = \ln\left(\lambda^{n} \cdot e^{-\lambda \sum_{i=1}^{n} x_i}\right) = n \ln(\lambda) + \left(-\lambda \cdot \sum_{i=1}^{n} x_i\right)$$

$$\frac{\partial \ell}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0 \rightarrow \hat{\lambda} = \frac{n}{\sum_i x_i}$$

$$\frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{n}{\lambda^2}$$

Bayesian estimator/estimate

The main idea is to utilize Bayes' theorem to include some form of prior information about parameters that is known into the estimation process.

$$\xi_{post}(\theta, \mathbf{x}) = \frac{L(\theta)\xi_{pri}(\theta)}{g_n(\mathbf{x})}.$$

To obtain an estimate from a ξ_{post}, different loss functions can be utilized. For more information see *DeGroot, M. H., & Schervish, M. J. (n.d.). Probability and Statistics (4th ed.). Pearson.*